

# IVANNA PYLYPENKO

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## EDUCATION

### University of Ottawa

Ottawa, ON

Honours Bachelor of Science in Computer Science (CO-OP), Third Year — GPA 9.7

Sept 2024 – Apr 2029(expected)

- **Coursework:** Computer Architecture I, Data Structures & Algorithms, Intro to Software Engineering, Discrete Structures, Programming Paradigms, Databases I, Professional Practice //Coursework in progress// *Programming Language Concepts, Software Design&Architecture, Design&Analysis of Algorithms.*

## EXPERIENCE

### Research Assistant

University of Ottawa/Department of Electrical Engineering and Computer Science

May 2026 – September 2026

- Developed end-to-end AI evaluation pipelines with automated evaluation workflows.
- Designed and managed evaluation processes combining human reviewers and AI-based evaluators.
- Engineered data pipelines to query, process, and prepare structured datasets.
- Analyzed evaluation results to identify model performance patterns, failure cases, and areas for improvement.

### Math Tutor

Mathnasium Learning Centre — 117 Centrepoin Dr, Nepean, ON

May 2025 – Now

- Tutored students from Grades 1–12, including advanced Grade 12 mathematics like functions, trigonometry, and calculus.
- Adapted explanations for different learning styles and support long-term skill growth.
- Communicated student progress clearly with families and staff.

## PROJECTS

### Hotel Management Web Application — Team Course Project (3 participants)

Java · Spring Boot · PostgreSQL · Web Development

- Developed a full-stack hotel management application using Java and Spring Boot, integrated with a PostgreSQL database.
- Implemented database operations for storing, retrieving, and updating hotel, room, and reservation data.
- Built backend functionality connecting application logic with persistent database storage.
- Collaborated with team members to integrate application components and troubleshoot backend functionality.

### OTAMS TutorApp — Team Course Project (6 members)

Java · OOP · UI Design · MVC · Git Collaboration

- Built a multi-role tutoring application for admins, teachers, and students.
- Designed and implemented major UI components, helper classes, and interaction logic and ensured smooth communication between UI and backend modules; resolved interaction bugs.
- Collaborated using Git branching, issue tracking, and structured development workflows.

### Paris Metro as a graph— Algorithms Project (6 members)

Java · DFS/BFS · Algorithm Design · File Processing

- Developed a clustering algorithm to analyze Paris Subway stations and repeatedly analyze them to find the cheapest/shortest routes.
- Implemented optimized DFS/BFS/ Dijkstra's traversal logic and recursive search techniques.
- Delivered accurate calculations and robust performance on large inputs.

### Simple CPU Design — Computer Architecture Project

Digital Logic · CPU Design · ALU & Control Flow

- Designed and simulated a functional, simple CPU featuring an ALU, register file, instruction decoder, and control unit.
- Demonstrated understanding of low-level architecture, execution cycles, and data flow.

## SKILLS

**Programming:** Java, Python, Go, SQL, Prolog, HTML, CSS.

**Frameworks & Tools:** Android, Firebase, REST APIs, Spring Boot, PostgreSQL, Git/GitHub, Android Studio, UML, VS Code, Eclipse.

**Software Development:** Object-Oriented Programming, MVC Architecture, Client-Server Architecture, REST APIs, Database Integration, File Parsing, Event-Driven Programming.

**Algorithms & Data Structures:** Data Structures, DFS, BFS, Dijkstra's Algorithm, Graph Algorithms, Recursion.

**AI & Data:** AI Model Evaluation, Data Pipelines, Database-Driven Applications, Human and AI-Based Evaluation.

**Languages:** English, Ukrainian, Russian.